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REFUTATION TEXT IN SCIENCE EDUCATION: A REVIEW OF TWO DECADES OF RESEARCH

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ABSTRACT. As people attempt to make sense of the world, they develop personal knowledge structures. These structures often contain misconceptions—inaccurate or incomplete information—that are highly resistant to change because existing knowledge networks must be restructured to accommodate counterintuitive information in a process known as conceptual change. Since textbooks are the dominant resource for science instruction in most classrooms, text-based methods of facilitating conceptual change need to be examined. Since the mid-1980 s, researchers have investigated the conceptual change potential of refutation text, a text structure that includes elements of argumentation and that has been described as one of the most effective text-based means for modifying readers' misconceptions. In this paper, twenty years of refutation text research in science and reading education is reviewed and then a secondary analysis of those results is conducted to explore developmental aspects of the efficacy of refutation text. Although a developmental relationship was not revealed, two decades of research indicate that reading refutation text rather than traditional expository text is more likely to result in conceptual change.

KEY WORDS: conceptual change, misconceptions, reading, refutation text, science education

Many people believe that an ostrich will bury its head in the sand when it is in danger. This is not true, however. If ostriches buried their heads, they would not be able to breathe! Ostrich chicks may hide from danger by lying with their necks stretched out along the ground. Adults may listen for sound with their heads near the ground, or they might run away. (Tippett, 2004)

After reading this passage, which consists of a text structure known as refutation text, Ian, a Grade 3 student, remarked, “*I used to think when ostriches were afraid they put their heads in the ground, but they don’t.*” His comment indicates the power of refutation text—reading a refutation text passage caused Ian to become aware of his misconception about ostriches and to change his thinking.

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INTRODUCTION

An international interest in using text to facilitate conceptual change is reflected in research exploring the use of refutation text and conceptual change text. The majority of published studies have been conducted in the United States (e.g., Hynd & Alvermann, 1986a; Maria & MacGinitie, 1987). However, research examining the use of refutation text or conceptual change text has also been conducted in Australia (Palmer, 2003), Canada (Kendeou & van den Broek, 2007), China (Chiu & Wong, 1995), Cyprus (Diakidoy, Kendeou, & Ioannides, 2002), Finland (Mikkilä-Erdmann, 2001), Italy (Mason & Gava, 2007), Taiwan (Tsai & Chou, 2002), and Turkey (Pinarbaşı, Canpolat, Bayrakçeken, & Geban, 2006; Tekkaya, 2003). In some studies, what is called conceptual change text *is* refutation text that engages, challenges, and remediates common misconceptions, while in other studies conceptual change text merely contains explanations for the phenomena about which misconceptions might be held rather than explicitly refuting those misconceptions. In this review, I exclude conceptual change text research unless the text can be identified as refutational.

A meta-analysis by Guzzetti, Snyder, & Glass (1992) identified refutation text, analogies, the learning cycle, and discrepant events as interventions that promoted conceptual change. However, although the efficacy of refutation text, or refutational text, is noted in current literature (e.g., Chambliss, 2002; Shanahan, 2004), textbooks typically contain very little refutation text (Guzzetti et al., 1992). In addition, an examination of 40 recent award winning children's science information tradebooks indicated that the refutation text structure is not widely used in current children's science information tradebooks (Tippett, 2009). The aims of this paper are to highlight the conceptual change potential of reading refutation text, as revealed by 20 years of research, to provide research-based evidence as encouragement for authors and publishers to include the refutation text structure in science information tradebooks and textbooks, and to encourage educators to utilize the refutation text structure during classroom science instruction.

The review begins with a description of refutation text and a brief discussion of children's science misconceptions, followed by a historical overview of theories of conceptual growth and change. The results of 20 years of refutation text research are then summarized and reviewed by research theme (Part 1), and re-examined in an effort to identify developmental trends in refutation text efficacy (Part 2). A discussion of current trends in refutation text research, recommendations for future research, and implications for educators and the publishing industry form the concluding remarks.

REFUTATION TEXT

Refutation text, sometimes called refutational text, is a text structure that challenges readers' misconceptions. Refutation text passages always contain two components: the statement of a commonly held misconception, and an explicit refutation of that misconception with an emphasis on the currently accepted scientific explanation (Guzzetti, 2000). A third component, a signal or cue that alerts the reader to the possibility of another conception, may also be present (Maria & MacGinitie, 1987). Figure 1 shows a refutation text passage and its three components.

MISCONCEPTIONS

The term *misconception* is frequently used to refer to a belief that conflicts with currently accepted scientific explanations. Other terms that are sometimes used include alternative belief, alternative conception, alternative framework, children's science, erroneous idea, inaccurate prior knowledge, intuitive conception, intuitive science, naïve conception, naïve theory, nonscientific idea, persistent pitfall, preconception, pre-instructional conception, and spontaneous reasoning (Chinn & Malhotra, 2002; Guzzetti, Snyder, Glass, & Gamas, 1993; Guzzetti, Williams, Skeels, & Wu, 1997; Wandersee, Mintzes, & Novak, 1994; White & Gunstone, 2008). The term chosen typically reflects the researchers' underlying assumptions about cognition and learning rather than indicating a different construct. For consistency and brevity, I use *misconception* unless citing specific research results, in which case the term adopted by the investigators is used.

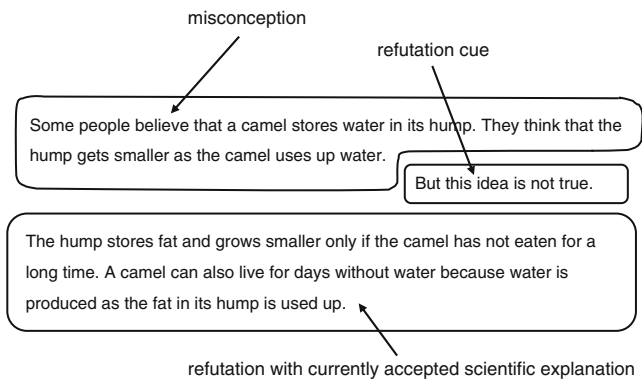


Figure 1. Refutation text components

Although misconceptions exist in almost every subject area, they seem especially prevalent in science (Maria, 2000). Most people possess science misconceptions, such as those shown in Table 1, which are ideas that have developed as the result of an individual's efforts to make sense of the world. These misconceptions are highly resistant to change (Dole & Smith, 1989; Maria). However, misconceptions *can* be transformed, during a process called conceptual change (Posner, Strike, Hewson, & Gertzog, 1982).

CONCEPTUAL CHANGE THEORIES: A BRIEF HISTORY OF THE LAST THIRTY YEARS

Most current theories of learning in science are constructivist-based: learners are seen to be active participants in the acquisition of skills and knowledge (Piaget, 1975/1977; Vygotsky, trans. 1986). Learners make sense of the world from the perspective of their developed knowledge structures (schema), which are adapted during assimilation or accommodation. Consideration of these two processes has provided the foundation for much conceptual change research: assimilation occurs when new information or concepts can be added without restructuring existing schema and can be considered *conceptual growth*; accommodation occurs when schema must be restructured to include new information or concepts and can be considered *conceptual change*.

The Piagetian notions of accommodation and assimilation and a recognition of the importance of prior knowledge contributed to the development of Posner et al.'s (1982) influential model of conceptual change (Scott, Asoko, & Leach, 2007). In this model, there are four possibilities when readers encounter new information, only one of which results in conceptual change. If the new information is not counterintuitive, it can be assimilated into existing prior knowledge, resulting in conceptual growth. If the new information is counterintuitive, however,

TABLE 1
Examples of common science misconceptions

All stars are the same size.
 The sun boils the sea to create water vapor.
 Objects float in water because they are lighter than water.
 The Earth is molten, except for its crust.
 A camel stores water in its hump.
 The heart beat is a reflex action.

Adapted from AIP Operation Physics Project, 1998; Maria and MacGinitie, 1987; Palmer, 2003; Sungur, Tekkaya, and Geban, 2001

there are three possibilities: the information might be rejected outright, the information might be memorized in a new knowledge structure that is not connected with or assimilated into prior knowledge structures, or prior knowledge structures might be restructured to accommodate the new information, resulting in conceptual change. Posner et al. (1982) also outlined four conditions that must be met if conceptual change is to occur: prior knowledge must be identified as inadequate to solve the problem, new information must be understandable (intelligible), new information must be useful in solving the current problem (plausible), and new information must appear useful in solving future problems (fruitful).

Early theories of conceptual change were based on the assumption that concepts—knowledge and explanations—were formed piecemeal, emerging through experience and added to as new information was acquired through further experience and through social transmission (White & Gunstone, 2008). Posner et al. (1982) considered conceptual change from this epistemological perspective, focusing on the evolution of concepts (Duit, Treagust, & Widodo, 2008). The process of conceptual change has also been considered from ontological and affective perspectives. An emphasis on how learners view the nature of scientific concepts is evident in the conceptual change theory proposed by Chi, Slotta, and de Leeuw (1994), in which the nature of ontological categories, science concepts, and naïve conceptions are assumed to influence learning. The role of emotions and social interactions is emphasized in the “hot cognition” model (Pintrich, Marx, & Boyle, 1993) based on an interactive relationship between the conditions for conceptual change and motivational factors such as goals, values, and personal interest.

Strike & Posner (1992) revised their earlier model to include learner motivation and the social contexts of the learner’s goals and clarified their interpretation of conceptual change as an interactive, developmental process that is based upon a learner’s conceptions and misconceptions. This revised model is more in line with the current view that conceptual change might be better considered from a multidimensional stance that integrates all three perspectives: epistemological, ontological, and affective (Duit et al., 2008).

The early conceptual change model proposed by Posner et al. (1982) provided the theoretical framework for early refutation text research (e.g., Hynd & Alvermann, 1986a, 1986b). Researchers in both science and reading education have examined the use of refutation text to promote conceptual change when readers possess science misconceptions. In the following section, a thematic review of twenty years of refutation text research is presented.

A REVIEW OF REFUTATION TEXT RESEARCH

The refutation text research results described in this review were located by conducting an electronic search of 66 academic databases including Academic Search Premier, ERIC, JSTOR, and PsycARTICLES for the phrases *refutation text* OR *refutational text* AND *science*. The bibliographies of meta-analyses and related articles (e.g., Guzzetti et al., 1992) were manually checked for relevant items. In addition, to locate any recent and as yet unpublished research, I conducted an online search of the programs and proceedings for three major North American and European associations: the American Educational Research Association (AERA) 2006–2009, the European Association for Research on Learning and Instruction (EARLI) 2007 and 2009, and the European Science Education Research Association (ESERA) 2007 and 2009. Finally, I eliminated duplicate items and items that did not report empirical results, which yielded 31 refutation text studies, listed by author in Table 2 (located in the Electronic Supplementary Materials system (ESM) of the journal), that are considered in this review.

Part 1: Themes in Refutation Text Research

The studies listed in Table 2 (available in the ESM of the journal) were categorized according to research focus in an open coding process that revealed six major themes—prior knowledge activation, format and structure of refutation text, other forms of refutation, refutation text and maintained conceptual change, reading processes and strategies used when reading refutation text, and factors that might contribute to effectiveness of refutation text—and I use those themes to organize Part 1 of this review of refutation text research. In Part 2 of the review, refutation text research results are analyzed to determine if the effectiveness of refutation text might be related to readers' grade level.

Prior Knowledge Activation Followed by Reading Refutation Text. Prior knowledge activation was a theme in seven of the studies included in this review. Prior knowledge can be activated in many ways, including questioning, giving notice that prior knowledge may be inaccurate, or presenting a demonstration. Hynd & Alvermann (1986a, 1986b) found that for developmental studies college students who held misconceptions, activation of inaccurate prior knowledge did not seem to have any significant effect on posttest scores, although physics students who had their misconceptions activated remembered fewer important ideas after

reading a text passage, regardless of text type. In a related study, Alvermann & Hague (1989) found that prior knowledge activation, when combined with an alert about possible inconsistencies in prior knowledge, did seem to result in better comprehension of counterintuitive science text. Marshall (1989), working with college students, found that the use of refutation text, followed by a demonstration that included the same information, was not as effective at producing conceptual change as was the demonstration followed by reading the refutation text. The difference in effectiveness may have been because prior knowledge was activated or perhaps even developed during the demonstration, although Marshall did not address this issue in her discussion of the results. However, she suggested that further research be undertaken to clarify and confirm the importance of the order of the combined activities. Guzzetti (1990) found that participating in a prereading activity intended to activate prior knowledge did not seem to lead to significant differences in students' posttest scores and that text type did not seem to influence posttest scores either. However, she noted that students who read refutation text after participating in the activity did score significantly better on application problems, indicating that the lack of significance might be related to the instrumentation rather than the interventions. Wang & Andre (1991) found significant differences on posttest measures when students' prior knowledge was activated through adjunct questions. Hynd, Alvermann, and Qian (1997) found that preservice teachers who viewed a demonstration before reading a text passage performed significantly better on posttests than students who only read the text.

In summary, the results of five of the seven studies examining the influence of prior knowledge activation in combination with refutation text indicate a positive effect on subsequent comprehension of related science concepts. Prior knowledge can be activated in many ways, including questioning, giving notice that prior knowledge may be inaccurate, or presenting a demonstration.

Format and Structure of Refutation Text. Six of the studies presented in this review explored the aspects of format and structure of refutation text. Gordon & Rennie (1987) suggested that when the same information is embedded in both expository and narrative formats, Grade 5 students may not find the information in a narrative format as believable as information presented in an expository format, and this suggestion was supported by the results of two refutation text studies. Mayer (1995) found that students in Kindergarten through Grade 3 did not acquire a significant amount of new information from a narrative soft expository text, while Alvermann,

Hynd & Qian (1995) found that Grade 9 students were more likely to learn concepts presented in an expository refutation text than those same concepts presented in a narrative refutation text. The authors postulated that readers may have been distracted by the storyline of the narrative refutation text. On the other hand, Maria & Johnson (1989) investigated the effectiveness of three text formats—traditional expository, refutational expository, and refutational soft expository in which science concepts were embedded in a story—and found that gifted students in Grades 5 and 7 who read a refutational soft expository text had higher comprehension scores. However, readability may also have been a factor in comprehension as the readability levels of the nonrefutational expository text and the soft expository refutational text were two grade levels lower than the readability of the refutational text.

Several refutation text studies have attempted to answer various questions about reader preference for text format or structure, because preference may contribute to motivation to read a particular text, and motivation, in turn, may contribute to reading comprehension (Duke & Pearson, 2002; Harp & Mayer, 1997). Maria & MacGinitie (1987) suggested that readers in Grades 5 and 6 had a preference for refutation text with the misconception component located at the beginning. A comparison of narrative, expository, and cartoon forms of refutation text indicated that students in Grades 11 and 12 preferred the expository format (Guzzetti et al., 1997). Hynd (2001) found that high school students preferred refutational texts to other kinds of text, and that they preferred an expository format over a narrative format. Mason, Gava, & Boldrin (2008) found that Grade 5 students preferred refutation text over traditional expository text. Reader preference may be a factor in the effectiveness of refutation text.

Research examining the most effective refutation text format has yielded inconsistent results, which may be due to the grade levels of the study participants, to researchers' differing ideas of narrative and soft expository text, or to confounding factors including readability. However, research investigating student preference has provided more consistent results: students prefer refutation text over traditional expository text, and they prefer an expository format over cartoons, narrative, or soft expository.

Other Forms of Refutation. Two of the studies presented in this review have examined the possibility of refutation presented in non-textual forms. Hynd, McWhorter, Phares & Suttles (1994) compared the effects of reading a refutation text passage with viewing a refutational

demonstration and with participating in peer discussion groups. The refutation text had the strongest effect on posttest scores, while the refutational discussion and demonstration did not significantly affect scores. Frède (2008) compared reading refutation text with participating in a refutation modelling activity, as well as with reading a traditional expository text. There were no significant differences between the two refutation interventions, which both resulted in significantly higher immediate posttest scores than reading a nonrefutation text. After one month, test scores for the refutation modelling group were significantly higher than both text groups, and the refutation text group scored higher than the expository text group.

Participating in refutational activities appears to be an intervention that merits further research. The activities described in these two studies include refutational demonstrations and refutational modelling, and other activities might include refutational writing or drawing—students creating their own refutational texts. Such refutational activities could be undertaken in conjunction with reading refutational text.

Refutation Text and Maintained Conceptual Change. Many of the studies included in this review have investigated maintained conceptual change through the use of delayed posttests. Tippet (2004) found that students in Grades 3 and 4 continued to show conceptual change six weeks after reading refutation text passages. Maria & Johnson (1989) found delayed effects a month after gifted students in Grades 5 and 7 had read both expository and soft expository refutation text. Mason & Gava (2007) found that after two months, Grade 8 students who had read refutational text passages were more likely to demonstrate maintained conceptual change. Hynd et al. (1994) found that two weeks after reading refutation text, students in Grades 9 and 10 continued to demonstrate changed conceptions. Hynd et al. (1997) found indications of maintained conceptual change two months after preservice teachers read refutation text passages, while Frède (2008) found that preservice teachers who read refutation text continued to show improved understanding on a posttest given one month after reading.

Many of the refutation text studies included in this review have indicated that the effects of reading refutation text may be maintained over time, and these maintained effects occur for participants in a wide range of grade levels. These findings are comparable to the results of a meta-analysis by Guzzetti et al. (1993), which indicated that reading refutation text was the only strategy that effectively facilitated long-term conceptual change.

Reading Processes and Strategies Used When Reading Refutation Text. Investigating the processes and strategies that are involved in reading refutation text passages is a recent theme in refutation text research, and is the theme of three studies in this review. Broughton, Sinatra, & Reynolds (2007) examined college students' attention allocation when reading refutation and nonrefutation text and found that there was a significant difference in time spent reading the passages, with participants spending *less* time reading the refutation text passage. However, they found that text type did not result in significant differences in posttest scores. Mikkilä-Erdmann, Penttinen, Anto, & Olkinuora (2008) used eye-tracking data to explore the processes involved when Grade 6 students read refutation or nonrefutation text passages. They found that increased conceptual change was related to time spent rereading previous sections of the passage, which they proposed was an indication of cognitive conflict. There were no significant differences due to text type. Kendeou & van den Broek (2007) found that university students with misconceptions used more conceptual change strategies, such as making inferences or contrasting information, than readers without misconceptions, but only when reading refutation text. They proposed that this difference in strategy use might be related to coactivation and integration of prior knowledge in a manner that facilitates identification of discrepancies.

The reading processes and strategies that students use when reading refutation text have only recently become a research focus. The results of these three studies indicate that readers do approach refutation and expository texts differently, and a closer examination of specific processes is a promising area for future research.

Factors that Might Affect Refutation Text Effectiveness. Six of the studies in this review shared the theme of investigating the factors that might contribute to the effectiveness of refutation text, including reader's commitment to misconceptions, reader's spatial ability, reader's epistemological beliefs, and the type of information contained. Dole & Niederhauser (1990) investigated how the degree of commitment to naïve conceptions would affect the likelihood of conceptual change for Grade 6 students. Using traditional, considerate, and refutation texts, they found that the level of commitment, determined by asking students how sure they were of their answers, had little effect on conceptual change learning from text. However, they noted that students who read refutation text were more likely to experience conceptual change than students who read traditional text. Chiu & Wong (1995) found that after reading

refutation text, Grade 9 students with high spatial ability tended to alter their mental models to become more scientifically accurate while students with low spatial ability were less likely to change their mental models. Qian (1995) used a refutation text passage to explore the effects of high school students' epistemological beliefs on conceptual change and found that students with naïve epistemological beliefs were less likely to experience conceptual change. Mason & Gava (2007) found that while Grade 8 students who read a refutation text passage were more likely than students who read a traditional expository text to experience conceptual change, those students with more advanced epistemological beliefs were even more likely to experience conceptual change. In a related study, Mason et al. (2008) found that after reading a refutation text passage, Grade 5 students with more advanced epistemological beliefs attained higher posttest scores than students with naïve beliefs about the nature of knowledge did. They also found a positive correlation between students' interest in a topic and conceptual change. The type of information presented in a passage may be a factor: Skopeliti & Vosniadou (2006) found that for Grade 3 students, reading a refutation text that included classificatory information was more likely than reading a text passage that simply refuted misconceptions to result in conceptual change.

Results from these six studies indicate that readers with more sophisticated epistemological beliefs and higher spatial ability may be more likely to experience conceptual change as a result of reading refutation text. The nature of the information contained in the refutation text passage may also affect the possibility of conceptual change. However, readers' level of commitment to beliefs does not appear to affect conceptual change associated with refutation text.

Twenty Years of Refutation Text Research—Results and Future Directions

In summary, reading refutation text is likely to be more effective when presented in an expository or soft expository format rather than in a narrative format (Alvermann et al., 1995; Gordon & Rennie, 1987; Maria & Johnson, 1989). Readers prefer refutation text over traditional expository text (Hynd, 2001; Mason, Hynd & Qian, 2008), and they prefer an expository format over narratives and cartoons (Guzzetti et al., 1997; Hynd). Although simply activating prior knowledge before reading refutation text may hinder conceptual change when that prior knowledge is inaccurate (Hynd & Alvermann, 1986a, 1986b), activating prior knowledge and adding a warning about possible inaccuracies may result in increased comprehension of related

science concepts presented in refutation text passages (Alvermann & Hague, 1989; Hynd et al., 1997; Marshall, 1989; Wang & Andre, 1991). Readers seem to allocate attention differently when presented with refutation text passages, and spend *less* time reading refutation text (Kendeou & van den Broek, 2007; Mikkilä-Erdmann et al., 2008). A reader's commitment to prior knowledge does not appear to influence the effectiveness of reading refutation text (Dole & Niederhauser, 1990). However, a reader's epistemological beliefs do seem to influence conceptual change (Mason & Gava, 2007; Mason et al., 2008; Qian, 1995). Finally, reading refutation text can result in maintained conceptual change (Tippett, 2004; Frède, 2008; Hynd et al., 1994; Maria & Johnson, 1989; Mason & Gava, 2007).

There are, of course, a number of areas that warrant further investigation, including replication studies and studies focusing on specific and as yet unexamined variables. For example, textual factors, including the effect of pictures accompanying text passages, the influence of the genre and format of refutation text passages, and the impact of readability level or concept loading of the text passage, could be further examined. Reader characteristics such as gender, cultural differences, preference for text format, and the degree of interest in a topic could also be further explored.

Part 2: A Grade Level Analysis of Refutation Text Research Results

Several meta-analyses have been conducted in the area of conceptual change, at least two of which focused on studies using text to promote that change (Guzzetti et al., 1992; Guzzetti et al., 1993). However, neither of these two meta-analyses focused exclusively on the use of refutation text or examined the possible relationship between grade level and effectiveness. To determine if there might be a developmental aspect to the effectiveness of refutation text, the results of two decades of refutation text research were examined according to the grade or education level of the participants (Rossman & Yore, 2009).

Studies were included in the grade level analysis only if the use of refutation text was directly compared to the use of nonrefutation text, as indicated in the right-hand column in Table 2 (located in the ESM of the journal). For example, if a study examined the use of refutation text in combination with another technique, it was included in this grade level analysis only if the results were compared with the results of using that same technique in conjunction with nonrefutation text. This criterion eliminated several studies indicating positive results for refutation text used in combination with other activities compared to nonrefutation text only (e.g., Alvermann et al., 1995; Hynd et al., 1997).

Unfortunately, effect sizes could not be used in this secondary analysis as differences in research design and in the ways in which results were reported preclude the use of a single method to calculate effect sizes, yielding such comparisons invalid. As an alternative, the 22 studies included in the analysis were grouped according to grade level to see if developmental trends might be revealed. Grouping the studies into the commonly used categories of elementary, secondary and university did not clearly indicate any relationships between the reader's grade or education level and the effectiveness of refutation text in promoting conceptual change. Instead, for each grade, the percent of conducted studies that found reading refutation text to be more effective than reading expository text was graphed, revealing three categories as shown in Figure 2, a visual representation of the effects of refutation text according to grade level.

For young students, Kindergarten to Grade 2, the results of a single study indicate that reading refutation text is no more likely than reading expository text to lead to conceptual change. The cumulative results of eleven studies indicate that the same is true for students in Grade 11 or higher. For students in Grades 3 through 10, however, the cumulative results of ten studies suggest that reading refutation text is indeed more likely than reading traditional expository text to result in conceptual change. Keeping in mind that reading refutation text was never found to

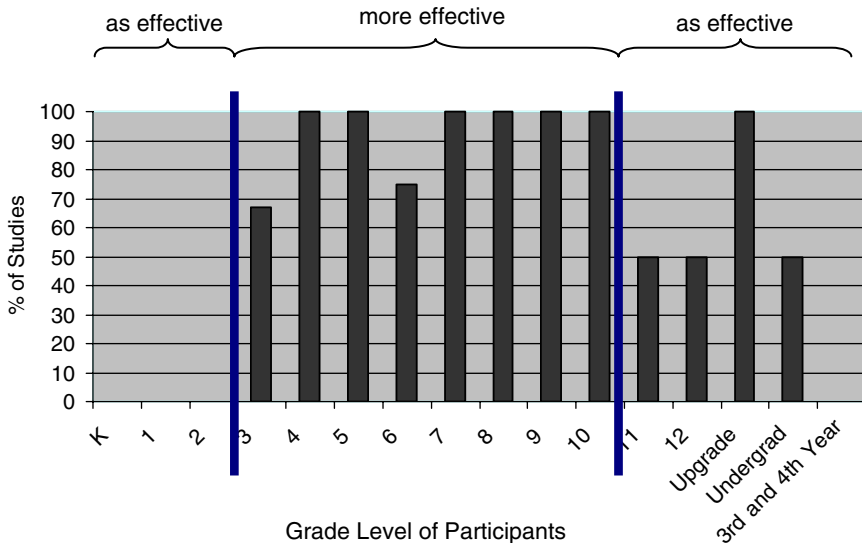


Figure 2. Percent of studies indicating that reading refutation text is more likely than reading expository text to result in conceptual change

be less effective than reading regular expository text at facilitating conceptual change, the results of 20 years of research conducted in both reading and science education are clear: reading refutation text can effectively promote conceptual change when readers possess science misconceptions, regardless of age or grade level.

The Power of Refutation Text

Why does reading refutation text seem to be an effective means of facilitating conceptual change? Several factors have been proposed as contributing to the power of refutation text, including a metacognitive awareness that prior knowledge is incorrect, the presentation of information in a plausible and coherent manner, and the nature of the information that is presented.

Metacognitive Awareness. Most descriptions of the conditions leading to conceptual change include a metacognitive awareness that one's prior or current knowledge is incorrect or inadequate (e.g., Posner et al., 1982), and this awareness might be caused by cognitive dissonance or cognitive conflict (Duit et al., 2008). It has been speculated that reading refutation text causes cognitive conflict, which can lead to conceptual change. Simulations of the process of reading refutation text have indicated that the proximity of the correct and incorrect conceptualizations may lead to an increased likelihood of simultaneous activation resulting in cognitive conflict (Kendeou & van den Broek, 2007). Because refutation text explicitly states a misconception and then refutes it, readers who hold that misconception are more likely to recognize that their prior knowledge is incorrect or inadequate. A traditional expository text passage does not contain an explicit cue that a current belief is inaccurate, and if a reader does not recognize that a current belief is incorrect, there is little reason to change that belief.

However, while cognitive conflict can be a major factor in conceptual change, it is not a necessary prerequisite, as in cases when current knowledge is lacking rather than incorrect. Nor does cognitive conflict always lead to the development of more accurate conceptions, since students may be aware of the incompatibility of new information but choose to ignore that information anyway (e.g., Chinn & Brewer, 1993) or develop new, yet still inaccurate mental models (Mikkilä-Erdmann et al., 2008). Other factors, such as the credibility and coherence of information and the nature of the misconception being addressed, may work in combination with a metacognitive awareness of the inadequacy of prior knowledge and lead to conceptual change.

Information that is Plausible and Coherent. Conceptual change is more likely to occur when anomalous information is presented in a believable or plausible manner (Chinn & Brewer, 1993). Refutation text can be viewed as an expert source, since both sides of an issue are presented, whereas traditional expository text presents only one side of an issue. The fact that both correct and incorrect information are contained in a text passage gives rise to the persuasive characteristics that are inherent in the refutation text structure (Hynd, 2001). A well-written refutation text passage presents information in a manner that is understandable, credible, and useful (Mason & Gava, 2007). The explicit reference to a misconception in refutation text may lead readers to find the correct information more acceptable than information presented without mention of alternate beliefs.

Kendeou & van den Broek (2007) found that reading times for target sentences were significantly lower when university students read refutation text passages, leading the authors to posit that the refutation text may be more coherent because of the explicit cue for integration of readers' prior knowledge and the about-to-be-presented information. In addition, the refutation text structure may provide readers with an advance organizer while at the same time activating prior knowledge and containing credible and coherent information (Broughton et al., 2007).

The Nature of the Misconception. The nature of the misconception being refuted may also contribute to the effectiveness of refutation text. Chi (2008) described three grain sizes of knowledge representations—single ideas, mental models, and categories—and suggested that faulty ideas might be revised through refutation, while flawed mental models would require multiple refutations, and robust misconceptions caused by incorrect categorization of information would need to be addressed at the categorical level. Refutation text may be more effective when misconceptions are single ideas or beliefs rather than complex concepts, as less change is required to accommodate new information (Chi).

IMPLICATIONS FOR EDUCATORS

This review of refutation text research has a number of implications for educators. First, students need explicit instruction about a variety of text formats, such as narrative or expository, and a variety of informational text structures, such as cause and effect or compare and contrast (e.g., Duke & Pearson, 2002). Instruction about refutation text would be appropriate during instruction about other text structures found in

informational texts. Second, for conceptual change to occur, students must be persuaded that their current conceptions are inaccurate (Chinn & Brewer, 1993). Therefore, teachers need to identify children's misconceptions and the nature of those misconceptions before designing classroom activities. Those activities might include reading refutation text specifically written to address the misconceptions held by a particular class or group of students. Teachers should also attempt to locate and use tradebooks and textbooks that contain refutation text. Although it would be difficult to address all misconceptions about a particular topic in a single book, the presence of some refutation text is more likely to promote conceptual change. Third, even when incorporating refutation text into classroom science instruction, teachers should continue to present concepts multiple times and in a variety of formats, particularly when students have flawed or robust mental models (Chi, 2008). Text alone, even if it is refutation text, is not enough to facilitate conceptual change for all students. Although refutation text appears effective for groups of students on average, it needs to be supplemented for some individuals, particularly those students with ineffective reading strategies (Guzzetti, 2000). Refutation text used in conjunction with other types of text, or with videos, demonstrations, hands-on experiments, and other activities, will increase the likelihood of conceptual change.

CONCLUDING REMARKS

While print texts, especially textbooks, are often viewed negatively by both teachers and students, these texts are the dominant method of science instruction (Newton, Newton, Blake, & Brown, 2002; Yore, Craig, & Maguire, 1998). The results from twenty years of research on the use of refutation text indicate that reading refutation text can lead to conceptual change, and that reading refutation text is more likely than reading traditional expository text to lead to the correction of misconceptions. Therefore, refutation text could (and should) be included in both tradebooks and textbooks. Unfortunately, a recent survey of award winning children's books indicates that the refutation text structure is relatively rare (Tippett, 2009). The power of refutation text as a conceptual change agent must be recognized by publishers, editors, and authors, so that the refutation text structure will appear more frequently in children's science information books. Educators should also be aware of the power of refutation text, so that they will choose books that incorporate refutation text, and perhaps even develop their own materials utilizing the refutation text structure.

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